Programming Assignment 2: Weighbridge Ticket Store

Data Structures & Algorithms Laboratory Handout (Medium)

Instructor: Bavith

Problem Overview

A weighbridge issues vehicle tickets throughout the day. Each ticket carries a ticket number, a vehicle registration number, a gross weight, and a tare weight. The net weight is calculated as gross minus tare.

Because the daily ticket count is known only when data arrives, static arrays cannot be used. Your program must dynamically allocate memory at runtime based on N, process records, compute totals and heaviest loads, and properly release memory before termination.

> **Input Validation Rule:** A count N of zero or below, or above 100,000, is strictly rejected with the output message: `INVALID COUNT`.

Ticket Structure & Helper Functions

Required Specifications

- **Ticket Structure:** Contains ticket number (int/long), vehicle registration (string), gross weight (float), and tare weight (float).
- **Helper Function 1:** Takes a pointer to a ticket and returns its net weight.
- **Helper Function 2:** Takes a pointer to a ticket and prints its formatted details.
- **Memory Management:** Allocate using `malloc()`/`calloc()` and release using `free()`.

Important Behaviors

Zero Net Weight: If a vehicle's gross weight equals its tare weight, its net weight is `0.00`. This is a legitimate operational event (e.g., an empty vehicle crossing the weighbridge) and must be printed rather than skipped.

Heaviest Load: Identify and reprint the ticket carrying the maximum net load.

Sample Test Cases

#	Input (stdin)	Expected Output (stdout)
1	3 9001 KA27AB1234 14250.50 4800.00 9002 KA25CD9876 21000.00 6250.75 9003 KA31EF4455 9800.25 3200.00	WEIGHBRIDGE DAY BOOK 9001 KA27AB1234 gross 14250.50 tare 4800.00 net 9450.50 9002 KA25CD9876 gross 21000.00 tare 6250.75 net 14749.25 9003 KA31EF4455 gross 9800.25 tare 3200.00 net 6600.25 Tickets: 3 Total net weight (kg): 30800.00 Heaviest net load: ticket 9002 KA25CD9876 14749.25
2	2 5501 KA28XY7788 8000.00 8000.00 5502 KA29ZZ0099 15500.75 5500.25	WEIGHBRIDGE DAY BOOK 5501 KA28XY7788 gross 8000.00 tare 8000.00 net 0.00 5502 KA29ZZ0099 gross 15500.75 tare 5500.25 net 10000.50 Tickets: 2 Total net weight (kg): 10000.50 Heaviest net load: ticket 5502 KA29ZZ0099 10000.50
3	0	INVALID COUNT

Submission Guidelines

Ensure your solution implements dynamic memory allocation correctly, includes the two required helper functions with pointer arguments, and frees allocated memory before exit.

> **Instructor Note:** Test your implementation with edge cases like zero net weight before final submission.
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> — Bavith